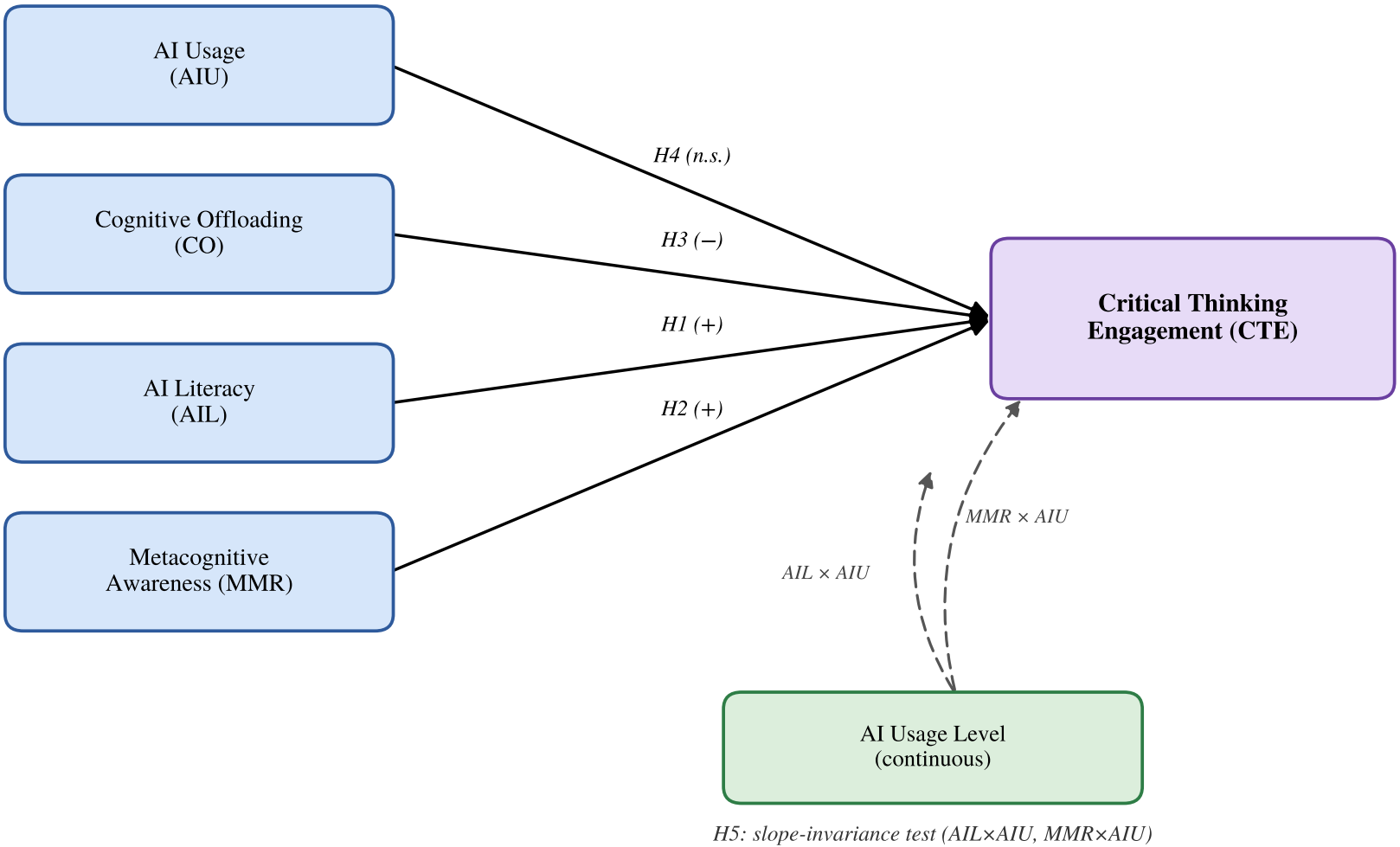
AI Usage
(AIU)
Cognitive Offloading
(CO)
AI Literacy
(AIL)
Metacognitive
Awareness (MMR)
H4 (n.s.)
H3 (−)
H1 (+)
H2 (+)
Critical Thinking
Engagement (CTE)
AIL × AIU
MMR × AIU
AI Usage Level
(continuous)
H5: slope-invariance test (AIL×AIU, MMR×AIU)